

1) ISP Router (ISP-RTR-01)

1.1) show running-config

ISP-RTR-01#show running-config

Building configuration...

Current configuration : 872 bytes

!

version 15.4

no service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname ISP-RTR-01

!

!

!

!

!

!

!

!

!

ip cef

no ipv6 cef

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

spanning-tree mode pvst

!

!

!

!

!

!

interface Loopback0

```
ip address 8.8.8.8 255.255.255.255
!
interface GigabitEthernet0/0/0
ip address 10.0.0.2 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/0/1
ip address 192.168.2.1 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 10.10.0.0 255.255.0.0 10.0.0.1
ip route 10.20.0.0 255.255.0.0 192.168.2.2
ip route 192.168.2.0 255.255.255.252 192.168.2.2
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end
```

ISP-RTR-01#

1.2) show ip route

ISP-RTR-01#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

8.0.0.0/32 is subnetted, 1 subnets
C 8.8.8.8/32 is directly connected, Loopback0
10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C 10.0.0.0/30 is directly connected, GigabitEthernet0/0/0
L 10.0.0.2/32 is directly connected, GigabitEthernet0/0/0
S 10.10.0.0/16 [1/0] via 10.0.0.1
S 10.20.0.0/16 [1/0] via 192.168.2.2
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.2.0/30 is directly connected, GigabitEthernet0/0/1
L 192.168.2.1/32 is directly connected, GigabitEthernet0/0/1

ISP-RTR-01#

2) CORP Router (HQ-EDGE-RTR-01)

2.1) show running-config

HQ-EDGE-RTR-01#show running-config
Building configuration...

Current configuration : 1955 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname HQ-EDGE-RTR-01

```
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
ip cef  
no ipv6 cef  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface Tunnel0  
ip address 10.255.0.1 255.255.255.252  
mtu 1476  
tunnel source GigabitEthernet0/0/0  
tunnel destination 192.168.2.2  
!  
!  
interface GigabitEthernet0/0/0  
ip address 10.0.0.1 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/0/0.40  
no ip address  
ip helper-address 172.16.0.10
```

```
!  
interface GigabitEthernet0/0/0.50  
no ip address  
ip helper-address 172.16.0.10  
!  
interface GigabitEthernet0/0/0.90  
no ip address  
ip helper-address 172.16.0.10  
!  
interface GigabitEthernet0/0/1  
no ip address  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/0/1.40  
encapsulation dot1Q 40  
ip address 10.10.0.2 255.255.254.0  
ip helper-address 172.16.0.10  
!  
interface GigabitEthernet0/0/1.50  
encapsulation dot1Q 50  
ip address 10.10.4.2 255.255.255.0  
ip helper-address 172.16.0.10  
!  
interface GigabitEthernet0/0/1.90  
encapsulation dot1Q 90  
ip address 10.10.2.2 255.255.254.0  
ip helper-address 172.16.0.10  
!  
interface GigabitEthernet0/0/2  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
router-id 1.1.1.1  
log-adjacency-changes  
passive-interface GigabitEthernet0/0/0  
network 10.10.0.0 0.0.1.255 area 0  
network 10.10.2.0 0.0.1.255 area 0  
network 10.10.4.0 0.0.0.255 area 0
```

```

network 10.10.5.0 0.0.0.255 area 0
network 10.10.6.0 0.0.0.127 area 0
network 10.10.6.192 0.0.0.63 area 0
network 10.10.7.0 0.0.0.31 area 0
network 172.16.0.0 0.0.0.255 area 0
network 10.255.0.0 0.0.0.3 area 0
default-information originate
!
ip classless
ip route 10.20.0.0 255.255.0.0 10.255.0.2
ip route 0.0.0.0 0.0.0.0 10.0.0.2
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end

```

```

HQ-EDGE-RTR-01#
HQ-EDGE-RTR-01#

```

2.2) show ip route

```
HQ-EDGE-RTR-01#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 10.0.0.2 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 15 subnets, 8 masks
 C 10.0.0.0/30 is directly connected, GigabitEthernet0/0/0
 L 10.0.0.1/32 is directly connected, GigabitEthernet0/0/0
 C 10.10.0.0/23 is directly connected, GigabitEthernet0/0/1.40
 L 10.10.0.2/32 is directly connected, GigabitEthernet0/0/1.40
 C 10.10.2.0/23 is directly connected, GigabitEthernet0/0/1.90
 L 10.10.2.2/32 is directly connected, GigabitEthernet0/0/1.90
 C 10.10.4.0/24 is directly connected, GigabitEthernet0/0/1.50
 L 10.10.4.2/32 is directly connected, GigabitEthernet0/0/1.50
 O 10.10.5.0/24 [110/2] via 10.10.0.1, 00:28:08, GigabitEthernet0/0/1.40
 [110/2] via 10.10.4.1, 00:28:08, GigabitEthernet0/0/1.50
 [110/2] via 10.10.2.1, 00:28:08, GigabitEthernet0/0/1.90
 O 10.10.6.0/25 [110/2] via 10.10.0.1, 00:28:08, GigabitEthernet0/0/1.40
 [110/2] via 10.10.4.1, 00:28:08, GigabitEthernet0/0/1.50
 [110/2] via 10.10.2.1, 00:28:08, GigabitEthernet0/0/1.90
 O 10.10.6.192/26 [110/2] via 10.10.0.1, 01:00:56, GigabitEthernet0/0/1.40
 [110/2] via 10.10.4.1, 01:00:56, GigabitEthernet0/0/1.50
 [110/2] via 10.10.2.1, 01:00:56, GigabitEthernet0/0/1.90
 O 10.10.7.0/27 [110/2] via 10.10.0.1, 00:28:08, GigabitEthernet0/0/1.40
 [110/2] via 10.10.4.1, 00:28:08, GigabitEthernet0/0/1.50
 [110/2] via 10.10.2.1, 00:28:08, GigabitEthernet0/0/1.90
 S 10.20.0.0/16 [1/0] via 10.255.0.2
 C 10.255.0.0/30 is directly connected, Tunnel0
 L 10.255.0.1/32 is directly connected, Tunnel0
 172.16.0.0/24 is subnetted, 1 subnets
 O 172.16.0.0/24 [110/2] via 10.10.0.1, 01:00:56, GigabitEthernet0/0/1.40
 [110/2] via 10.10.4.1, 01:00:56, GigabitEthernet0/0/1.50
 [110/2] via 10.10.2.1, 01:00:56, GigabitEthernet0/0/1.90
 S* 0.0.0.0/0 [1/0] via 10.0.0.2

HQ-EDGE-RTR-01#

HQ-EDGE-RTR-01#

HQ-EDGE-RTR-01#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.2	1	FULL/DR	00:00:33	10.10.0.1	GigabitEthernet0/0/1.40
1.1.1.2	1	FULL/DR	00:00:33	10.10.2.1	GigabitEthernet0/0/1.90
1.1.1.2	1	FULL/DR	00:00:33	10.10.4.1	GigabitEthernet0/0/1.50

HQ-EDGE-RTR-01#show ip ospf interface brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gig 1	0	10.10.0.2/255.255.254.0	1	BDR	0/0		
Gig 1	0	10.10.4.2/255.255.255.0	1	BDR	0/0		

```
Gig 1 0 10.10.2.2/255.255.254.0 1 BDR 0/0  
Tun0 1 0 10.255.0.1/255.255.255.252 1000 POINT 0/0
```

```
HQ-EDGE-RTR-01#show interfaces trunk
```

```
HQ-EDGE-RTR-01#show ip nat translations  
HQ-EDGE-RTR-01#
```

3) Core L3 Switch (HQ-CORE-SW1)

```
4) Current configuration : 3259 bytes  
5) !  
6) version 16.3.2  
7) no service timestamps log datetime msec  
8) no service timestamps debug datetime msec  
9) no service password-encryption  
10) !  
11) hostname HQ_CORE_SW1  
12) !  
13) !  
14) no profinet  
15) !  
16) !  
17) !  
18) !  
19) !  
20) !  
21) no ip cef  
22) ip routing  
23) !  
24) no ipv6 cef  
25) !  
26) !  
27) !  
28) !  
29) !  
30) !  
31) !  
32) !  
33) !  
34) !  
35) !
```


36) !
37) !
38) !
39) spanning-tree mode pvst
40) !
41) !
42) !
43) !
44) !
45) !
46) interface GigabitEthernet1/0/1
47) switchport trunk native vlan 99
48) switchport mode trunk
49) !
50) interface GigabitEthernet1/0/2
51) switchport trunk native vlan 99
52) switchport mode trunk
53) spanning-tree portfast trunk
54) !
55) interface GigabitEthernet1/0/3
56) switchport trunk native vlan 99
57) switchport trunk allowed vlan 10,20,30,40,50,60,80,90,99
58) switchport mode trunk
59) !
60) interface GigabitEthernet1/0/4
61) switchport trunk native vlan 99
62) switchport mode trunk
63) !
64) interface GigabitEthernet1/0/5
65) switchport trunk native vlan 99
66) switchport trunk allowed vlan 10,20,30,40,50,60,80,90,99
67) switchport mode trunk
68) !
69) interface GigabitEthernet1/0/6
70) switchport trunk native vlan 99
71) switchport mode trunk
72) !
73) interface GigabitEthernet1/0/7
74) !
75) interface GigabitEthernet1/0/8
76) !
77) interface GigabitEthernet1/0/9
78) !
79) interface GigabitEthernet1/0/10
80) !
81) interface GigabitEthernet1/0/11

```
82)!
83)interface GigabitEthernet1/0/12
84)!
85)interface GigabitEthernet1/0/13
86)!
87)interface GigabitEthernet1/0/14
88)!
89)interface GigabitEthernet1/0/15
90)!
91)interface GigabitEthernet1/0/16
92)!
93)interface GigabitEthernet1/0/17
94)!
95)interface GigabitEthernet1/0/18
96)!
97)interface GigabitEthernet1/0/19
98)!
99)interface GigabitEthernet1/0/20
100)    !
101)    interface GigabitEthernet1/0/21
102)    !
103)    interface GigabitEthernet1/0/22
104)    !
105)    interface GigabitEthernet1/0/23
106)    !
107)    interface GigabitEthernet1/0/24
108)    !
109)    interface GigabitEthernet1/1/1
110)    !
111)    interface GigabitEthernet1/1/2
112)    !
113)    interface GigabitEthernet1/1/3
114)    !
115)    interface GigabitEthernet1/1/4
116)    !
117)    interface Vlan1
118)    no ip address
119)    shutdown
120)    !
121)    interface Vlan10
122)    mac-address 0005.5e41.a201
123)    ip address 10.10.6.193 255.255.255.192
124)    !
125)    interface Vlan20
126)    mac-address 0005.5e41.a202
127)    ip address 10.10.5.1 255.255.255.0
```

```
128) ip helper-address 172.16.0.10
129) !
130) interface Vlan30
131) mac-address 0005.5e41.a203
132) ip address 10.10.6.1 255.255.255.128
133) ip helper-address 172.16.0.10
134) !
135) interface Vlan40
136) mac-address 0005.5e41.a204
137) ip address 10.10.0.1 255.255.254.0
138) ip helper-address 172.16.0.10
139) !
140) interface Vlan50
141) mac-address 0005.5e41.a205
142) ip address 10.10.4.1 255.255.255.0
143) ip helper-address 172.16.0.10
144) !
145) interface Vlan60
146) mac-address 0005.5e41.a206
147) ip address 10.10.7.1 255.255.255.224
148) ip helper-address 172.16.0.10
149) !
150) interface Vlan80
151) mac-address 0005.5e41.a207
152) ip address 172.16.0.1 255.255.255.0
153) ip helper-address 172.16.0.10
154) !
155) interface Vlan90
156) mac-address 0005.5e41.a208
157) ip address 10.10.2.1 255.255.254.0
158) ip helper-address 172.16.0.10
159) !
160) interface Vlan99
161) mac-address 0005.5e41.a209
162) ip address 192.168.1.1 255.255.255.0
163) !
164) router ospf 1
165) router-id 1.1.1.2
166) log-adjacency-changes
167) network 10.10.0.0 0.0.1.255 area 0
168) network 10.10.2.0 0.0.1.255 area 0
169) network 10.10.4.0 0.0.0.255 area 0
170) network 10.10.5.0 0.0.0.255 area 0
171) network 10.10.6.0 0.0.0.127 area 0
172) network 10.10.6.192 0.0.0.63 area 0
173) network 10.10.7.0 0.0.0.31 area 0
```

```

174)    network 172.16.0.0 0.0.0.255 area 0
175)    !
176)    ip classless
177)    ip route 0.0.0.0 0.0.0.0 192.168.1.2
178)    ip route 10.20.0.0 255.255.0.0 10.10.0.2
179)    !
180)    ip flow-export version 9
181)    !
182)    !
183)    !
184)    !
185)    !
186)    !
187)    !
188)    line con 0
189)    !
190)    line aux 0
191)    !
192)    line vty 0 4
193)    login
194)    !
195)    !
196)    !
197)    !
198)    end
199)
200)
201)    HQ_CORE_SW1#
202)    HQ_CORE_SW1#
203)    HQ_CORE_SW1#

```

Show vlan brief

HQ_CORE_SW1#show vlan brief

VLAN Name Status Ports

```

-----
1 default active Gig1/0/7, Gig1/0/8, Gig1/0/9, Gig1/0/10
Gig1/0/11, Gig1/0/12, Gig1/0/13, Gig1/0/14
Gig1/0/15, Gig1/0/16, Gig1/0/17, Gig1/0/18
Gig1/0/19, Gig1/0/20, Gig1/0/21, Gig1/0/22
Gig1/0/23, Gig1/0/24, Gig1/1/1, Gig1/1/2
Gig1/1/3, Gig1/1/4
10 MGMT active
20 FINANCE active
30 HR active

```

40 SALES active
50 ENGINEERING active
60 EXEC active
80 SERVERS active
90 VOIP active
99 NATIVE active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
HQ_CORE_SW1#

HQ_CORE_SW1#show interfaces trunk
Port Mode Encapsulation Status Native vlan
Gig1/0/1 on 802.1q trunking 99
Gig1/0/2 on 802.1q trunking 99
Gig1/0/3 on 802.1q trunking 99
Gig1/0/4 on 802.1q trunking 99
Gig1/0/5 on 802.1q trunking 99
Gig1/0/6 on 802.1q trunking 99

Port Vlans allowed on trunk
Gig1/0/1 1-1005
Gig1/0/2 1-1005
Gig1/0/3 10,20,30,40,50,60,80,90,99
Gig1/0/4 1-1005
Gig1/0/5 10,20,30,40,50,60,80,90,99
Gig1/0/6 1-1005

Port Vlans allowed and active in management domain
Gig1/0/1 1,10,20,30,40,50,60,80,90,99
Gig1/0/2 1,10,20,30,40,50,60,80,90,99
Gig1/0/3 10,20,30,40,50,60,80,90,99
Gig1/0/4 1,10,20,30,40,50,60,80,90,99
Gig1/0/5 10,20,30,40,50,60,80,90,99
Gig1/0/6 1,10,20,30,40,50,60,80,90,99

Port Vlans in spanning tree forwarding state and not pruned
Gig1/0/1 1,10,20,30,40,50,60,80,90,99
Gig1/0/2 none
Gig1/0/3 10,20,30,40,50,60,80,90,99
Gig1/0/4 1,10,20,30,40,50,60,80,90,99
Gig1/0/5 10,20,30,40,50,60,80,90,99

Gig1/0/6 1,10,20,30,40,50,60,80,90,99

HQ_CORE_SW1

HQ_CORE_SW1#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1/0/1	unassigned	YES	unset	up	up
GigabitEthernet1/0/2	unassigned	YES	unset	up	up
GigabitEthernet1/0/3	unassigned	YES	unset	up	up
GigabitEthernet1/0/4	unassigned	YES	unset	up	up
GigabitEthernet1/0/5	unassigned	YES	unset	up	up
GigabitEthernet1/0/6	unassigned	YES	unset	up	up
GigabitEthernet1/0/7	unassigned	YES	unset	down	down
GigabitEthernet1/0/8	unassigned	YES	unset	down	down
GigabitEthernet1/0/9	unassigned	YES	unset	down	down
GigabitEthernet1/0/10	unassigned	YES	unset	down	down
GigabitEthernet1/0/11	unassigned	YES	unset	down	down
GigabitEthernet1/0/12	unassigned	YES	unset	down	down
GigabitEthernet1/0/13	unassigned	YES	unset	down	down
GigabitEthernet1/0/14	unassigned	YES	unset	down	down
GigabitEthernet1/0/15	unassigned	YES	unset	down	down
GigabitEthernet1/0/16	unassigned	YES	unset	down	down
GigabitEthernet1/0/17	unassigned	YES	unset	down	down
GigabitEthernet1/0/18	unassigned	YES	unset	down	down
GigabitEthernet1/0/19	unassigned	YES	unset	down	down
GigabitEthernet1/0/20	unassigned	YES	unset	down	down
GigabitEthernet1/0/21	unassigned	YES	unset	down	down
GigabitEthernet1/0/22	unassigned	YES	unset	down	down
GigabitEthernet1/0/23	unassigned	YES	unset	down	down
GigabitEthernet1/0/24	unassigned	YES	unset	down	down
GigabitEthernet1/1/1	unassigned	YES	unset	down	down
GigabitEthernet1/1/2	unassigned	YES	unset	down	down
GigabitEthernet1/1/3	unassigned	YES	unset	down	down
GigabitEthernet1/1/4	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset administratively	down	down
Vlan10	10.10.6.193	YES	manual	up	up
Vlan20	10.10.5.1	YES	manual	up	up
Vlan30	10.10.6.1	YES	manual	up	up
Vlan40	10.10.0.1	YES	manual	up	up
Vlan50	10.10.4.1	YES	manual	up	up
Vlan60	10.10.7.1	YES	manual	up	up
Vlan80	172.16.0.1	YES	manual	up	up
Vlan90	10.10.2.1	YES	manual	up	up

Vlan99 192.168.1.1 YES manual up up
HQ_CORE_SW1#

HQ_CORE_SW1#show vtp status
VTP Version capable : 1 to 2
VTP version running : 2
VTP Domain Name : itinfra2026.local
VTP Pruning Mode : Disabled
VTP Traps Generation : Disabled
Device ID : 0002.17ED.0C00
Configuration last modified by 0.0.0.0 at 3-3-93 01:51:09
Local updater ID is 10.10.6.193 on interface V110 (lowest numbered VLAN interface found)

Feature VLAN :

VTP Operating Mode : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 14
Configuration Revision : 76
MD5 digest : 0xFB 0xD8 0x66 0x08 0x57 0xDB 0x08 0x47
0x7C 0x82 0x68 0x6F 0x66 0xFF 0x57 0x3F
HQ_CORE_SW1#

HQ_CORE_SW1#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port

Number of channel-groups in use: 0
Number of aggregators: 0

Group Port-channel Protocol Ports

-----+-----+-----+-----
HQ_CORE_SW1#

HQ_CORE_SW1#show ip ospf neighbor

```
Neighbor ID Pri State Dead Time Address Interface
1.1.1.3 1 FULL/DR 00:00:39 10.10.5.2 Vlan20
1.1.1.3 1 FULL/DR 00:00:39 10.10.6.2 Vlan30
1.1.1.1 1 FULL/BDR 00:00:39 10.10.0.2 Vlan40
1.1.1.1 1 FULL/BDR 00:00:39 10.10.4.2 Vlan50
1.1.1.3 1 FULL/DR 00:00:39 10.10.7.2 Vlan60
1.1.1.3 1 FULL/DR 00:00:39 172.16.0.2 Vlan80
1.1.1.1 1 FULL/BDR 00:00:39 10.10.2.2 Vlan90
1.1.1.3 1 FULL/DR 00:00:39 10.10.6.194 Vlan10
HQ_CORE_SW1#
```

HQ_CORE_SW1#show spanning-tree summary

Switch is in pvst mode

Root bridge for:

Extended system ID is enabled

Portfast Default is disabled

PortFast BPDU Guard Default is disabled

Portfast BPDU Filter Default is disabled

Loopguard Default is disabled

EtherChannel misconfig guard is disabled

UplinkFast is disabled

BackboneFast is disabled

Configured Pathcost method used is short

Name Blocking Listening Learning Forwarding STP Active

VLAN0001 1 0 0 4 5
VLAN0010 1 0 0 5 6
VLAN0020 1 0 0 5 6
VLAN0030 1 0 0 5 6
VLAN0040 1 0 0 5 6
VLAN0050 1 0 0 5 6
VLAN0060 1 0 0 5 6
VLAN0080 1 0 0 5 6
VLAN0090 1 0 0 5 6

VLAN0099 1 0 0 5 6

10 vlans 10 0 0 49 59

HQ_CORE_SW1#

HQ-CORE-SW2

HQ_CORE_SW2#show running-config

Building configuration...

Current configuration : 2595 bytes

!

version 16.3.2

no service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname HQ_CORE_SW2

!

!

no profinet

!

!

!

!

!

!

no ip cef

ip routing

!

no ipv6 cef

!

!

!

!

!

!

!

!

!

!

!

!

```
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet1/0/1  
switchport trunk native vlan 99  
switchport mode trunk  
!  
interface GigabitEthernet1/0/2  
switchport trunk native vlan 99  
switchport mode trunk  
!  
interface GigabitEthernet1/0/3  
switchport trunk native vlan 99  
switchport mode trunk  
!  
interface GigabitEthernet1/0/4  
switchport trunk native vlan 99  
switchport mode trunk  
!  
interface GigabitEthernet1/0/5  
switchport trunk native vlan 99  
switchport mode trunk  
!  
interface GigabitEthernet1/0/6  
!  
interface GigabitEthernet1/0/7  
!  
interface GigabitEthernet1/0/8  
!  
interface GigabitEthernet1/0/9  
!  
interface GigabitEthernet1/0/10  
!  
interface GigabitEthernet1/0/11  
!  
interface GigabitEthernet1/0/12  
!  
interface GigabitEthernet1/0/13  
!  
interface GigabitEthernet1/0/14
```

```
!  
interface GigabitEthernet1/0/15  
!  
interface GigabitEthernet1/0/16  
!  
interface GigabitEthernet1/0/17  
!  
interface GigabitEthernet1/0/18  
!  
interface GigabitEthernet1/0/19  
!  
interface GigabitEthernet1/0/20  
!  
interface GigabitEthernet1/0/21  
!  
interface GigabitEthernet1/0/22  
!  
interface GigabitEthernet1/0/23  
!  
interface GigabitEthernet1/0/24  
!  
interface GigabitEthernet1/1/1  
!  
interface GigabitEthernet1/1/2  
!  
interface GigabitEthernet1/1/3  
!  
interface GigabitEthernet1/1/4  
!  
interface Vlan1  
no ip address  
shutdown  
!  
interface Vlan10  
mac-address 0060.3ece.3e01  
ip address 10.10.6.194 255.255.255.192  
!  
interface Vlan20  
mac-address 0060.3ece.3e02  
ip address 10.10.5.2 255.255.255.0  
!  
interface Vlan30  
mac-address 0060.3ece.3e03  
ip address 10.10.6.2 255.255.255.128  
!  
interface Vlan60
```

```
mac-address 0060.3ece.3e04
ip address 10.10.7.2 255.255.255.224
!
interface Vlan80
mac-address 0060.3ece.3e05
ip address 172.16.0.2 255.255.255.0
!
interface Vlan99
mac-address 0060.3ece.3e06
ip address 192.168.1.2 255.255.255.0
!
router ospf 1
router-id 1.1.1.3
log-adjacency-changes
network 10.10.0.0 0.0.1.255 area 0
network 10.10.2.0 0.0.1.255 area 0
network 10.10.4.0 0.0.0.255 area 0
network 10.10.5.0 0.0.0.255 area 0
network 10.10.6.0 0.0.0.127 area 0
network 10.10.6.192 0.0.0.63 area 0
network 10.10.7.0 0.0.0.31 area 0
network 172.16.0.0 0.0.0.255 area 0
!
ip classless
ip route 10.20.0.0 255.255.0.0 10.10.5.1
ip route 10.20.0.0 255.255.0.0 10.10.0.2
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
!
end
```

HQ_CORE_SW2#

HQ_CORE_SW2#show vlan brief

VLAN Name Status Ports

1 default active Gig1/0/6, Gig1/0/7, Gig1/0/8, Gig1/0/9
Gig1/0/10, Gig1/0/11, Gig1/0/12, Gig1/0/13
Gig1/0/14, Gig1/0/15, Gig1/0/16, Gig1/0/17
Gig1/0/18, Gig1/0/19, Gig1/0/20, Gig1/0/21
Gig1/0/22, Gig1/0/23, Gig1/0/24, Gig1/1/1
Gig1/1/2, Gig1/1/3, Gig1/1/4
10 MGMT active
20 FINANCE active
30 HR active
40 SALES active
50 ENGINEERING active
60 EXEC active
80 SERVERS active
90 VOIP active
99 NATIVE active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
HQ_CORE_SW2#

HQ_CORE_SW2#show interfaces trunk

Port Mode Encapsulation Status Native vlan

Gig1/0/1 on 802.1q trunking 99
Gig1/0/2 on 802.1q trunking 99
Gig1/0/3 on 802.1q trunking 99
Gig1/0/4 on 802.1q trunking 99
Gig1/0/5 on 802.1q trunking 99

Port Vlans allowed on trunk

Gig1/0/1 1-1005
Gig1/0/2 1-1005
Gig1/0/3 1-1005
Gig1/0/4 1-1005
Gig1/0/5 1-1005

Port Vlans allowed and active in management domain

Gig1/0/1 1,10,20,30,40,50,60,80,90,99
Gig1/0/2 1,10,20,30,40,50,60,80,90,99
Gig1/0/3 1,10,20,30,40,50,60,80,90,99
Gig1/0/4 1,10,20,30,40,50,60,80,90,99
Gig1/0/5 1,10,20,30,40,50,60,80,90,99

Port Vlans in spanning tree forwarding state and not pruned

Gig1/0/1 1,10,20,30,40,50,60,80,90,99
Gig1/0/2 1,10,20,30,40,50,60,80,90,99
Gig1/0/3 1,10,20,30,40,50,60,80,90,99
Gig1/0/4 1,10,20,30,40,50,60,80,90,99
Gig1/0/5 1,10,20,30,40,50,60,80,90,99

HQ_CORE_SW2#

HQ_CORE_SW2#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1/0/1	unassigned	YES	unset	up	up
GigabitEthernet1/0/2	unassigned	YES	unset	up	up
GigabitEthernet1/0/3	unassigned	YES	unset	up	up
GigabitEthernet1/0/4	unassigned	YES	unset	up	up
GigabitEthernet1/0/5	unassigned	YES	unset	up	up
GigabitEthernet1/0/6	unassigned	YES	unset	down	down
GigabitEthernet1/0/7	unassigned	YES	unset	down	down
GigabitEthernet1/0/8	unassigned	YES	unset	down	down
GigabitEthernet1/0/9	unassigned	YES	unset	down	down
GigabitEthernet1/0/10	unassigned	YES	unset	down	down
GigabitEthernet1/0/11	unassigned	YES	unset	down	down
GigabitEthernet1/0/12	unassigned	YES	unset	down	down
GigabitEthernet1/0/13	unassigned	YES	unset	down	down
GigabitEthernet1/0/14	unassigned	YES	unset	down	down
GigabitEthernet1/0/15	unassigned	YES	unset	down	down
GigabitEthernet1/0/16	unassigned	YES	unset	down	down
GigabitEthernet1/0/17	unassigned	YES	unset	down	down
GigabitEthernet1/0/18	unassigned	YES	unset	down	down
GigabitEthernet1/0/19	unassigned	YES	unset	down	down
GigabitEthernet1/0/20	unassigned	YES	unset	down	down
GigabitEthernet1/0/21	unassigned	YES	unset	down	down
GigabitEthernet1/0/22	unassigned	YES	unset	down	down
GigabitEthernet1/0/23	unassigned	YES	unset	down	down
GigabitEthernet1/0/24	unassigned	YES	unset	down	down
GigabitEthernet1/1/1	unassigned	YES	unset	down	down
GigabitEthernet1/1/2	unassigned	YES	unset	down	down
GigabitEthernet1/1/3	unassigned	YES	unset	down	down

```
GigabitEthernet1/1/4 unassigned YES unset down down
Vlan1 unassigned YES unset administratively down down
Vlan10 10.10.6.194 YES manual up up
Vlan20 10.10.5.2 YES manual up up
Vlan30 10.10.6.2 YES manual up up
Vlan60 10.10.7.2 YES manual up up
Vlan80 172.16.0.2 YES manual up up
Vlan99 192.168.1.2 YES manual up up
HQ_CORE_SW2#
```

```
HQ_CORE_SW2#show vtp status
VTP Version capable : 1 to 2
VTP version running : 2
VTP Domain Name : itinfra2026.local
VTP Pruning Mode : Disabled
VTP Traps Generation : Disabled
Device ID : 0007.EC0B.0200
Configuration last modified by 0.0.0.0 at 3-3-93 01:51:09
Local updater ID is 10.10.6.194 on interface Vl10 (lowest numbered VLAN interface found)
```

Feature VLAN :

```
-----
VTP Operating Mode : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 14
Configuration Revision : 76
MD5 digest : 0xFB 0xD8 0x66 0x08 0x57 0xDB 0x08 0x47
0x7C 0x82 0x68 0x6F 0x66 0xFF 0x57 0x3F
HQ_CORE_SW2#
```

```
HQ_CORE_SW2#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
```

Number of channel-groups in use: 0

Number of aggregators: 0

Group Port-channel Protocol Ports

-----+-----+-----+-----

HQ_CORE_SW2#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.2	1	FULL/BDR	00:00:37	10.10.6.193	Vlan10
1.1.1.2	1	FULL/BDR	00:00:37	10.10.5.1	Vlan20
1.1.1.2	1	FULL/BDR	00:00:37	10.10.6.1	Vlan30
1.1.1.2	1	FULL/BDR	00:00:37	10.10.7.1	Vlan60
1.1.1.2	1	FULL/BDR	00:00:37	172.16.0.1	Vlan80

HQ_CORE_SW2#show spanning-tree summary

Switch is in pvst mode

Root bridge for:

Extended system ID is enabled

Portfast Default is disabled

PortFast BPDU Guard Default is disabled

Portfast BPDU Filter Default is disabled

Loopguard Default is disabled

EtherChannel misconfig guard is disabled

UplinkFast is disabled

BackboneFast is disabled

Configured Pathcost method used is short

Name	Blocking	Listening	Learning	Forwarding	STP	Active
------	----------	-----------	----------	------------	-----	--------

VLAN0001 0 0 0 5 5

VLAN0010 0 0 0 5 5

VLAN0020 0 0 0 5 5

VLAN0030 0 0 0 5 5

VLAN0040 0 0 0 5 5

VLAN0050 0 0 0 5 5

VLAN0060 0 0 0 5 5

VLAN0080 0 0 0 5 5

VLAN0090 0 0 0 5 5

VLAN0099 0 0 0 5 5

10 vlans 0 0 0 50 50

HQ_CORE_SW2#

Distribution Switches (HQ-DIST-SW1, SW2, SW3, SW4)

HQ-DIST-SW1

Switch#show running-config
Building configuration...

Current configuration : 1628 bytes

!

version 16.3.2

no service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname Switch

!

!

no profinet

!

!

!

!

!

!

no ip cef

no ipv6 cef

!

!

!

!

!

!

!

!

!

!

!

!

!

!

spanning-tree mode pvst

!

```
!  
!  
!  
!  
!  
interface GigabitEthernet1/0/1  
switchport trunk native vlan 99  
switchport mode trunk  
!  
interface GigabitEthernet1/0/2  
switchport trunk native vlan 99  
switchport mode trunk  
!  
interface GigabitEthernet1/0/3  
switchport trunk native vlan 99  
switchport trunk allowed vlan 80,99  
switchport mode trunk  
!  
interface GigabitEthernet1/0/4  
!  
interface GigabitEthernet1/0/5  
switchport trunk native vlan 99  
!  
interface GigabitEthernet1/0/6  
switchport trunk native vlan 99  
!  
interface GigabitEthernet1/0/7  
!  
interface GigabitEthernet1/0/8  
!  
interface GigabitEthernet1/0/9  
!  
interface GigabitEthernet1/0/10  
!  
interface GigabitEthernet1/0/11  
!  
interface GigabitEthernet1/0/12  
!  
interface GigabitEthernet1/0/13  
!  
interface GigabitEthernet1/0/14  
!  
interface GigabitEthernet1/0/15  
!  
interface GigabitEthernet1/0/16  
!
```

```
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
no ip address
shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
```

```
!  
!  
end
```

```
Switch#
```

```
Switch#show vlan brief
```

```
VLAN Name Status Ports
```

```
-----  
1 default active Gig1/0/4, Gig1/0/7, Gig1/0/8, Gig1/0/9  
Gig1/0/10, Gig1/0/11, Gig1/0/12, Gig1/0/13  
Gig1/0/14, Gig1/0/15, Gig1/0/16, Gig1/0/17  
Gig1/0/18, Gig1/0/19, Gig1/0/20, Gig1/0/21  
Gig1/0/22, Gig1/0/23, Gig1/0/24, Gig1/1/1  
Gig1/1/2, Gig1/1/3, Gig1/1/4
```

```
10 MGMT active
```

```
20 FINANCE active
```

```
30 HR active
```

```
40 SALES active
```

```
50 ENGINEERING active
```

```
60 EXEC active
```

```
80 SERVERS active
```

```
90 VOIP active
```

```
99 NATIVE active
```

```
1002 fddi-default active
```

```
1003 token-ring-default active
```

```
1004 fddinet-default active
```

```
1005 trnet-default active
```

```
Switch#show interfaces trunk
```

```
Port Mode Encapsulation Status Native vlan
```

```
Gig1/0/1 on 802.1q trunking 99
```

```
Gig1/0/2 on 802.1q trunking 99
```

```
Gig1/0/3 on 802.1q trunking 99
```

```
Gig1/0/5 auto n-802.1q trunking 99
```

```
Gig1/0/6 auto n-802.1q trunking 99
```

```
Port Vlans allowed on trunk
```

```
Gig1/0/1 1-1005
```

```
Gig1/0/2 1-1005
```

```
Gig1/0/3 80,99
```

```
Gig1/0/5 1-1005
```

```
Gig1/0/6 1-1005
```

Port Vlans allowed and active in management domain

Gig1/0/1 1,10,20,30,40,50,60,80,90,99

Gig1/0/2 1,10,20,30,40,50,60,80,90,99

Gig1/0/3 80,99

Gig1/0/5 1,10,20,30,40,50,60,80,90,99

Gig1/0/6 1,10,20,30,40,50,60,80,90,99

Port Vlans in spanning tree forwarding state and not pruned

Gig1/0/1 1,10,20,30,40,50,60,80,90,99

Gig1/0/2 1,10,20,30,40,50,60,80,90,99

Gig1/0/3 80,99

Gig1/0/5 1,10,20,30,40,50,60,80,90,99

Gig1/0/6 1,10,20,30,40,50,60,80,90,99

Switch#

Switch#

Switch#

Switch#

Switch#

Switch#

Switch#

Switch#show vtp status

VTP Version capable : 1 to 2

VTP version running : 2

VTP Domain Name : itinfra2026.local

VTP Pruning Mode : Disabled

VTP Traps Generation : Disabled

Device ID : 0001.43C1.B400

Configuration last modified by 0.0.0.0 at 3-3-93 11:44:43

Feature VLAN :

VTP Operating Mode : Client

Maximum VLANs supported locally : 1005

Number of existing VLANs : 14

Configuration Revision : 94

MD5 digest : 0x09 0x4A 0x5F 0x03 0x2B 0xE8 0x13 0x31

0x4B 0x81 0xD7 0xDD 0x27 0xFC 0xE1 0x1A

Switch#show etherchannel summary

Flags: D - down P - in port-channel

I - stand-alone s - suspended

H - Hot-standby (LACP only)

R - Layer3 S - Layer2

U - in use f - failed to allocate aggregator

u - unsuitable for bundling

w - waiting to be aggregated

d - default port

Number of channel-groups in use: 0

Number of aggregators: 0

Group Port-channel Protocol Ports

-----+-----+-----+-----

Switch#

HQ_DIST_SW2

Switch#show running-config

Building configuration...

Current configuration : 1616 bytes

!

version 16.3.2

no service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname Switch

!

!

no profinet

!

!

!

!

!

!

no ip cef

no ipv6 cef

!

!

!

!

!

!

!

```
!  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet1/0/1  
switchport trunk native vlan 99  
switchport mode trunk  
!  
interface GigabitEthernet1/0/2  
switchport trunk native vlan 99  
switchport mode trunk  
!  
interface GigabitEthernet1/0/3  
!  
interface GigabitEthernet1/0/4  
!  
interface GigabitEthernet1/0/5  
switchport trunk native vlan 99  
!  
interface GigabitEthernet1/0/6  
switchport trunk native vlan 99  
switchport trunk allowed vlan 10,20,30,40,50,60,80,90,99  
switchport mode trunk  
!  
interface GigabitEthernet1/0/7  
!  
interface GigabitEthernet1/0/8  
!  
interface GigabitEthernet1/0/9  
!  
interface GigabitEthernet1/0/10  
!  
interface GigabitEthernet1/0/11  
!  
interface GigabitEthernet1/0/12  
!
```

```
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
no ip address
shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
```



```
line con 0
!  
line aux 0
!  
line vty 0 4
login
!  
!  
!  
!  
end
```

Switch#

Switch#show vlan brief

VLAN Name Status Ports

```
-----  
1 default active Gig1/0/3, Gig1/0/4, Gig1/0/7, Gig1/0/8  
Gig1/0/9, Gig1/0/10, Gig1/0/11, Gig1/0/12  
Gig1/0/13, Gig1/0/14, Gig1/0/15, Gig1/0/16  
Gig1/0/17, Gig1/0/18, Gig1/0/19, Gig1/0/20  
Gig1/0/21, Gig1/0/22, Gig1/0/23, Gig1/0/24  
Gig1/1/1, Gig1/1/2, Gig1/1/3, Gig1/1/4  
10 MGMT active  
20 FINANCE active  
30 HR active  
40 SALES active  
50 ENGINEERING active  
60 EXEC active  
80 SERVERS active  
90 VOIP active  
99 NATIVE active  
1002 fddi-default active  
1003 token-ring-default active  
1004 fddinet-default active  
1005 trnet-default active
```

Switch#show interfaces trunk

Port Mode Encapsulation Status Native vlan

```
Gig1/0/1 on 802.1q trunking 99  
Gig1/0/2 on 802.1q trunking 99  
Gig1/0/5 auto n-802.1q trunking 99  
Gig1/0/6 on 802.1q trunking 99
```

Port Vlans allowed on trunk

Gig1/0/1 1-1005

Gig1/0/2 1-1005

Gig1/0/5 1-1005

Gig1/0/6 10,20,30,40,50,60,80,90,99

Port Vlans allowed and active in management domain

Gig1/0/1 1,10,20,30,40,50,60,80,90,99

Gig1/0/2 1,10,20,30,40,50,60,80,90,99

Gig1/0/5 1,10,20,30,40,50,60,80,90,99

Gig1/0/6 10,20,30,40,50,60,80,90,99

Port Vlans in spanning tree forwarding state and not pruned

Gig1/0/1 1,10,20,30,40,50,60,80,90,99

Gig1/0/2 1

Gig1/0/5 1,10,20,30,40,50,60,80,90,99

Gig1/0/6 10,20,30,40,50,60,80,90,99

Switch#

Switch#show vtp status

VTP Version capable : 1 to 2

VTP version running : 2

VTP Domain Name : itinfra2026.local

VTP Pruning Mode : Disabled

VTP Traps Generation : Disabled

Device ID : 000B.BEA1.6B00

Configuration last modified by 0.0.0.0 at 3-3-93 01:51:09

Feature VLAN :

VTP Operating Mode : Client

Maximum VLANs supported locally : 1005

Number of existing VLANs : 14

Configuration Revision : 76

MD5 digest : 0xFB 0xD8 0x66 0x08 0x57 0xDB 0x08 0x47

0x7C 0x82 0x68 0x6F 0x66 0xFF 0x57 0x3F

Switch#show etherchannel summary

Flags: D - down P - in port-channel

I - stand-alone s - suspended

H - Hot-standby (LACP only)

R - Layer3 S - Layer2

U - in use f - failed to allocate aggregator

u - unsuitable for bundling

w - waiting to be aggregated

d - default port

Number of channel-groups in use: 0
Number of aggregators: 0

Group Port-channel Protocol Ports

-----+-----+-----+-----

Switch#

HQ_DIST_SW3

HQ-DIST-SW3#show running-config
Building configuration...

Current configuration : 1658 bytes

!

version 16.3.2

no service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname HQ-DIST-SW3

!

!

no profinet

!

!

!

!

!

!

no ip cef

no ipv6 cef

!

!

!

!

!

!

!

!

!

!

!

```
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet1/0/1  
switchport trunk native vlan 99  
switchport mode trunk  
!  
interface GigabitEthernet1/0/2  
switchport trunk native vlan 99  
switchport mode trunk  
!  
interface GigabitEthernet1/0/3  
!  
interface GigabitEthernet1/0/4  
!  
interface GigabitEthernet1/0/5  
switchport trunk native vlan 99  
!  
interface GigabitEthernet1/0/6  
switchport trunk native vlan 99  
!  
interface GigabitEthernet1/0/7  
!  
interface GigabitEthernet1/0/8  
!  
interface GigabitEthernet1/0/9  
!  
interface GigabitEthernet1/0/10  
!  
interface GigabitEthernet1/0/11  
!  
interface GigabitEthernet1/0/12  
!  
interface GigabitEthernet1/0/13  
!  
interface GigabitEthernet1/0/14  
!  
interface GigabitEthernet1/0/15  
!
```

```
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
no ip address
shutdown
!
interface Vlan10
mac-address 0005.5e98.c501
ip address 10.10.6.196 255.255.255.192
!
ip default-gateway 10.10.6.193
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
```

```
!  
line aux 0  
!  
line vty 0 4  
login  
!  
!  
!  
!  
end
```

HQ-DIST-SW3#

HQ-DIST-SW3#show vlan brief

VLAN Name Status Ports

```
-----  
1 default active Gig1/0/3, Gig1/0/4, Gig1/0/7, Gig1/0/8  
Gig1/0/9, Gig1/0/10, Gig1/0/11, Gig1/0/12  
Gig1/0/13, Gig1/0/14, Gig1/0/15, Gig1/0/16  
Gig1/0/17, Gig1/0/18, Gig1/0/19, Gig1/0/20  
Gig1/0/21, Gig1/0/22, Gig1/0/23, Gig1/0/24  
Gig1/1/1, Gig1/1/2, Gig1/1/3, Gig1/1/4  
10 MGMT active  
20 FINANCE active  
30 HR active  
40 SALES active  
50 ENGINEERING active  
60 EXEC active  
80 SERVERS active  
90 VOIP active  
99 NATIVE active  
1002 fddi-default active  
1003 token-ring-default active  
1004 fddinet-default active  
1005 trnet-default active
```

HQ-DIST-SW3#show interfaces trunk

Port Mode Encapsulation Status Native vlan

```
Gig1/0/1 on 802.1q trunking 99  
Gig1/0/2 on 802.1q trunking 99  
Gig1/0/5 auto n-802.1q trunking 99  
Gig1/0/6 auto n-802.1q trunking 99
```

Port Vlans allowed on trunk

Gig1/0/1 1-1005
Gig1/0/2 1-1005
Gig1/0/5 1-1005
Gig1/0/6 1-1005

Port Vlans allowed and active in management domain

Gig1/0/1 1,10,20,30,40,50,60,80,90,99
Gig1/0/2 1,10,20,30,40,50,60,80,90,99
Gig1/0/5 1,10,20,30,40,50,60,80,90,99
Gig1/0/6 1,10,20,30,40,50,60,80,90,99

Port Vlans in spanning tree forwarding state and not pruned

Gig1/0/1 1,10,20,30,40,50,60,80,90,99
Gig1/0/2 none
Gig1/0/5 1,10,20,30,40,50,60,80,90,99
Gig1/0/6 1,10,20,30,40,50,60,80,90,99

HQ-DIST-SW3#

HQ-DIST-SW3#show vtp status

VTP Version capable : 1 to 2

VTP version running : 2

VTP Domain Name : itinfra2026.local

VTP Pruning Mode : Disabled

VTP Traps Generation : Disabled

Device ID : 0060.3E38.5900

Configuration last modified by 0.0.0.0 at 3-3-93 01:51:09

Feature VLAN :

VTP Operating Mode : Client

Maximum VLANs supported locally : 1005

Number of existing VLANs : 14

Configuration Revision : 76

MD5 digest : 0xFB 0xD8 0x66 0x08 0x57 0xDB 0x08 0x47

0x7C 0x82 0x68 0x6F 0x66 0xFF 0x57 0x3F

HQ-DIST-SW3#show etherchannel summary

Flags: D - down P - in port-channel

I - stand-alone s - suspended

H - Hot-standby (LACP only)

R - Layer3 S - Layer2

U - in use f - failed to allocate aggregator

u - unsuitable for bundling

w - waiting to be aggregated

d - default port

Number of channel-groups in use: 0
Number of aggregators: 0

Group Port-channel Protocol Ports

-----+-----+-----+-----

HQ-DIST-SW3#

HQ-DIST-SW4

HQ-DIST-SW4#show running-config
Building configuration...

Current configuration : 1570 bytes

!

version 16.3.2

no service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname HQ-DIST-SW4

!

!

no profinet

!

!

!

!

!

!

no ip cef

no ipv6 cef

!

!

!

!

!

!

!

!

!

!

!

!

!


```
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet1/0/1  
switchport trunk native vlan 99  
switchport mode trunk  
!  
interface GigabitEthernet1/0/2  
switchport trunk native vlan 99  
switchport mode trunk  
spanning-tree portfast trunk  
!  
interface GigabitEthernet1/0/3  
!  
interface GigabitEthernet1/0/4  
!  
interface GigabitEthernet1/0/5  
switchport trunk native vlan 99  
!  
interface GigabitEthernet1/0/6  
switchport trunk native vlan 99  
!  
interface GigabitEthernet1/0/7  
!  
interface GigabitEthernet1/0/8  
!  
interface GigabitEthernet1/0/9  
!  
interface GigabitEthernet1/0/10  
!  
interface GigabitEthernet1/0/11  
!  
interface GigabitEthernet1/0/12  
!  
interface GigabitEthernet1/0/13  
!  
interface GigabitEthernet1/0/14  
!  
interface GigabitEthernet1/0/15  
!  
interface GigabitEthernet1/0/16
```

```
!  
interface GigabitEthernet1/0/17  
!  
interface GigabitEthernet1/0/18  
!  
interface GigabitEthernet1/0/19  
!  
interface GigabitEthernet1/0/20  
!  
interface GigabitEthernet1/0/21  
!  
interface GigabitEthernet1/0/22  
!  
interface GigabitEthernet1/0/23  
!  
interface GigabitEthernet1/0/24  
!  
interface GigabitEthernet1/1/1  
!  
interface GigabitEthernet1/1/2  
!  
interface GigabitEthernet1/1/3  
!  
interface GigabitEthernet1/1/4  
!  
interface Vlan1  
no ip address  
shutdown  
!  
ip classless  
!  
ip flow-export version 9  
!  
!  
!  
!  
!  
!  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
login  
!
```

```
!  
!  
!  
end
```

HQ-DIST-SW4#

HQ-DIST-SW4#show vlan brief

VLAN Name Status Ports

1 default active Gig1/0/3, Gig1/0/4, Gig1/0/7, Gig1/0/8
Gig1/0/9, Gig1/0/10, Gig1/0/11, Gig1/0/12
Gig1/0/13, Gig1/0/14, Gig1/0/15, Gig1/0/16
Gig1/0/17, Gig1/0/18, Gig1/0/19, Gig1/0/20
Gig1/0/21, Gig1/0/22, Gig1/0/23, Gig1/0/24
Gig1/1/1, Gig1/1/2, Gig1/1/3, Gig1/1/4

10 MGMT active

20 FINANCE active

30 HR active

40 SALES active

50 ENGINEERING active

60 EXEC active

80 SERVERS active

90 VOIP active

99 NATIVE active

1002 fddi-default active

1003 token-ring-default active

1004 fddinet-default active

1005 trnet-default active

HQ-DIST-SW4#show interfaces trunk

Port Mode Encapsulation Status Native vlan

Gig1/0/1 on 802.1q trunking 99

Gig1/0/2 on 802.1q trunking 99

Gig1/0/5 auto n-802.1q trunking 99

Gig1/0/6 auto n-802.1q trunking 99

Port Vlans allowed on trunk

Gig1/0/1 1-1005

Gig1/0/2 1-1005

Gig1/0/5 1-1005

Gig1/0/6 1-1005

Port Vlans allowed and active in management domain

Gig1/0/1 1,10,20,30,40,50,60,80,90,99
Gig1/0/2 1,10,20,30,40,50,60,80,90,99
Gig1/0/5 1,10,20,30,40,50,60,80,90,99
Gig1/0/6 1,10,20,30,40,50,60,80,90,99

Port Vlans in spanning tree forwarding state and not pruned

Gig1/0/1 1,10,20,30,40,50,60,80,90,99
Gig1/0/2 none
Gig1/0/5 1,10,20,30,40,50,60,80,90,99
Gig1/0/6 1,10,20,30,40,50,60,80,90,99

HQ-DIST-SW4#

HQ-DIST-SW4#

HQ-DIST-SW4#show vtp status

VTP Version capable : 1 to 2

VTP version running : 2

VTP Domain Name : itinfra2026.local

VTP Pruning Mode : Disabled

VTP Traps Generation : Disabled

Device ID : 0004.9A13.AB00

Configuration last modified by 0.0.0.0 at 3-3-93 01:51:09

Feature VLAN :

VTP Operating Mode : Client

Maximum VLANs supported locally : 1005

Number of existing VLANs : 14

Configuration Revision : 76

MD5 digest : 0xFB 0xD8 0x66 0x08 0x57 0xDB 0x08 0x47

0x7C 0x82 0x68 0x6F 0x66 0xFF 0x57 0x3F

HQ-DIST-SW4#show etherchannel summary

Flags: D - down P - in port-channel

I - stand-alone s - suspended

H - Hot-standby (LACP only)

R - Layer3 S - Layer2

U - in use f - failed to allocate aggregator

u - unsuitable for bundling

w - waiting to be aggregated

d - default port

Number of channel-groups in use: 0

Number of aggregators: 0

Group Port-channel Protocol Ports

-----+-----+-----+-----

HQ-DIST-SW4#

Access Switches (representatives ACC-SW6, ACC-SW1)

ACC_SW6

HQ-ACC-SW6#show running-config
Building configuration...

Current configuration : 2397 bytes

```
!  
version 15.0  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname HQ-ACC-SW6  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
interface FastEthernet0/1  
switchport access vlan 90  
switchport mode access  
spanning-tree portfast  
spanning-tree bpduguard enable  
!  
interface FastEthernet0/2  
description UNUSED PORT - DISABLED  
shutdown  
!  
interface FastEthernet0/3  
description UNUSED PORT - DISABLED
```

```
shutdown
!
interface FastEthernet0/4
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/5
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/6
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/7
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/8
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/9
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/10
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/11
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/12
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/13
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/14
description UNUSED PORT - DISABLED
shutdown
!
```

```
interface FastEthernet0/15
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/16
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/17
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/18
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/19
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/20
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/21
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/22
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/23
description UNUSED PORT - DISABLED
shutdown
!
interface FastEthernet0/24
description UNUSED PORT - DISABLED
shutdown
!
interface GigabitEthernet0/1
switchport trunk native vlan 99
switchport mode trunk
!
interface GigabitEthernet0/2
!
```

```

interface Vlan1
no ip address
shutdown
!
interface Vlan10
ip address 10.10.6.195 255.255.255.192
!
ip default-gateway 10.10.6.196
!
!
!
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
!
!
!
end

```

```

HQ-ACC-SW6#
HQ-ACC-SW6#show vlan brief

```

VLAN Name Status Ports

```

-----
1 default active Fa0/2, Fa0/3, Fa0/4, Fa0/5
Fa0/6, Fa0/7, Fa0/8, Fa0/9
Fa0/10, Fa0/11, Fa0/12, Fa0/13
Fa0/14, Fa0/15, Fa0/16, Fa0/17
Fa0/18, Fa0/19, Fa0/20, Fa0/21
Fa0/22, Fa0/23, Fa0/24, Gig0/2
10 MGMT active
20 FINANCE active
30 HR active
40 SALES active
50 ENGINEERING active
60 EXEC active
80 SERVERS active
90 VOIP active Fa0/1
99 NATIVE active
1002 fddi-default active

```


1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
HQ-ACC-SW6#

HQ-ACC-SW6#show spanning-tree
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0001.4226.98AC
Cost 16
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0090.0C07.995D
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0010
Spanning tree enabled protocol ieee
Root ID Priority 32778
Address 0001.4226.98AC
Cost 16
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32778 (priority 32768 sys-id-ext 10)
Address 0090.0C07.995D
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0020
Spanning tree enabled protocol ieee
Root ID Priority 32788
Address 0001.4226.98AC
Cost 16
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32788 (priority 32768 sys-id-ext 20)
Address 0090.0C07.995D
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0030
Spanning tree enabled protocol ieee
Root ID Priority 32798
Address 0001.4226.98AC
Cost 16
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32798 (priority 32768 sys-id-ext 30)
Address 0090.0C07.995D
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0040
Spanning tree enabled protocol ieee
Root ID Priority 32808
Address 0001.4226.98AC
Cost 16
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32808 (priority 32768 sys-id-ext 40)
Address 0090.0C07.995D
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0050
Spanning tree enabled protocol ieee

Root ID Priority 32818
Address 0001.4226.98AC
Cost 16
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32818 (priority 32768 sys-id-ext 50)
Address 0090.0C07.995D
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0060
Spanning tree enabled protocol ieee
Root ID Priority 32828
Address 0001.4226.98AC
Cost 16
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32828 (priority 32768 sys-id-ext 60)
Address 0090.0C07.995D
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0080
Spanning tree enabled protocol ieee
Root ID Priority 32848
Address 0001.4226.98AC
Cost 16
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32848 (priority 32768 sys-id-ext 80)
Address 0090.0C07.995D
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0090

Spanning tree enabled protocol ieee

Root ID Priority 32858

Address 0001.4226.98AC

Cost 16

Port 25(GigabitEthernet0/1)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32858 (priority 32768 sys-id-ext 90)

Address 0090.0C07.995D

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Fa0/1 Desg FWD 19 128.1 P2p

Gi0/1 Root FWD 4 128.25 P2p

VLAN0099

Spanning tree enabled protocol ieee

Root ID Priority 32867

Address 0001.4226.98AC

Cost 16

Port 25(GigabitEthernet0/1)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32867 (priority 32768 sys-id-ext 99)

Address 0090.0C07.995D

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

HQ-ACC-SW6#

HQ-ACC-SW6#

HQ-ACC-SW6#

HQ-ACC-SW6#

HQ-ACC-SW6#

HQ-ACC-SW6#

HQ-ACC-SW6#

HQ-ACC-SW6#

```
HQ-ACC-SW6#
HQ-ACC-SW6#
HQ-ACC-SW6#show vtp status
VTP Version capable : 1 to 2
VTP version running : 2
VTP Domain Name : itinfra2026.local
VTP Pruning Mode : Disabled
VTP Traps Generation : Disabled
Device ID : 0002.165A.8500
Configuration last modified by 0.0.0.0 at 3-3-93 01:51:09
```

Feature VLAN :

```
-----
VTP Operating Mode : Client
Maximum VLANs supported locally : 255
Number of existing VLANs : 14
Configuration Revision : 76
MD5 digest : 0xFB 0xD8 0x66 0x08 0x57 0xDB 0x08 0x47
0x7C 0x82 0x68 0x6F 0x66 0xFF 0x57 0x3F
HQ-ACC-SW6#
```

ACC_SW1:

```
Switch#show running-config
Building configuration...
```

```
Current configuration : 1350 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport access vlan 20
switchport mode access
spanning-tree portfast
```

```
spanning-tree bpduguard enable
!
interface FastEthernet0/2
switchport access vlan 20
switchport mode access
spanning-tree portfast
spanning-tree bpduguard enable
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
```

```

interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
switchport trunk native vlan 99
switchport mode trunk
!
interface GigabitEthernet0/2
!
interface Vlan1
no ip address
shutdown
!
!
!
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
!
!
!
end

```

Switch#

Switch#show vlan brief

VLAN Name Status Ports

```

-----
1 default active Fa0/3, Fa0/4, Fa0/5, Fa0/6
Fa0/7, Fa0/8, Fa0/9, Fa0/10
Fa0/11, Fa0/12, Fa0/13, Fa0/14
Fa0/15, Fa0/16, Fa0/17, Fa0/18
Fa0/19, Fa0/20, Fa0/21, Fa0/22
Fa0/23, Fa0/24, Gig0/2
10 MGMT active
20 FINANCE active Fa0/1, Fa0/2
30 HR active

```

40 SALES active
50 ENGINEERING active
60 EXEC active
80 SERVERS active
90 VOIP active
99 NATIVE active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#

Switch#show spanning-tree
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0001.4226.98AC
Cost 4
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0001.9724.532E
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0010
Spanning tree enabled protocol ieee
Root ID Priority 32778
Address 0001.4226.98AC
Cost 4
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32778 (priority 32768 sys-id-ext 10)
Address 0001.9724.532E
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0020

Spanning tree enabled protocol ieee

Root ID Priority 32788

Address 0001.4226.98AC

Cost 4

Port 25(GigabitEthernet0/1)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32788 (priority 32768 sys-id-ext 20)

Address 0001.9724.532E

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Fa0/1 Desg FWD 19 128.1 P2p

Fa0/2 Desg FWD 19 128.2 P2p

Gi0/1 Root FWD 4 128.25 P2p

VLAN0030

Spanning tree enabled protocol ieee

Root ID Priority 32798

Address 0001.4226.98AC

Cost 4

Port 25(GigabitEthernet0/1)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32798 (priority 32768 sys-id-ext 30)

Address 0001.9724.532E

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0040

Spanning tree enabled protocol ieee

Root ID Priority 32808

Address 0001.4226.98AC

Cost 4

Port 25(GigabitEthernet0/1)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32808 (priority 32768 sys-id-ext 40)

Address 0001.9724.532E

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0050

Spanning tree enabled protocol ieee

Root ID Priority 32818

Address 0001.4226.98AC

Cost 4

Port 25(GigabitEthernet0/1)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32818 (priority 32768 sys-id-ext 50)

Address 0001.9724.532E

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0060

Spanning tree enabled protocol ieee

Root ID Priority 32828

Address 0001.4226.98AC

Cost 4

Port 25(GigabitEthernet0/1)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32828 (priority 32768 sys-id-ext 60)

Address 0001.9724.532E

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0080

Spanning tree enabled protocol ieee

Root ID Priority 32848

Address 0001.4226.98AC

Cost 4

Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32848 (priority 32768 sys-id-ext 80)
Address 0001.9724.532E
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0090
Spanning tree enabled protocol ieee
Root ID Priority 32858
Address 0001.4226.98AC
Cost 4
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32858 (priority 32768 sys-id-ext 90)
Address 0001.9724.532E
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

VLAN0099
Spanning tree enabled protocol ieee
Root ID Priority 32867
Address 0001.4226.98AC
Cost 4
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32867 (priority 32768 sys-id-ext 99)
Address 0001.9724.532E
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Gi0/1 Root FWD 4 128.25 P2p

```
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#show vtp status
VTP Version capable : 1 to 2
VTP version running : 2
VTP Domain Name : itinfra2026.local
VTP Pruning Mode : Disabled
VTP Traps Generation : Disabled
Device ID : 00E0.A301.B900
Configuration last modified by 0.0.0.0 at 3-3-93 11:44:43
```

Feature VLAN :

```
VTP Operating Mode : Client
Maximum VLANs supported locally : 255
Number of existing VLANs : 14
Configuration Revision : 94
MD5 digest : 0x09 0x4A 0x5F 0x03 0x2B 0xE8 0x13 0x31
0x4B 0x81 0xD7 0xDD 0x27 0xFC 0xE1 0x1A
Switch#
```

Branch Router (BRANCH-RTR-01)

```
BRANCH-RTR-01#show running-config
Building configuration...
```

```
Current configuration : 1216 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

```
!  
hostname BRANCH-RTR-01  
!  
!  
!  
!  
!  
!  
!  
!  
ip cef  
no ipv6 cef  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
interface Tunnel0  
ip address 10.255.0.2 255.255.255.252  
mtu 1476  
tunnel source GigabitEthernet0/0/1  
tunnel destination 10.0.0.1  
!  
!  
interface GigabitEthernet0/0/0  
no ip address  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/0/0.70
```

```
encapsulation dot1Q 70
ip address 10.20.0.1 255.255.254.0
!
interface GigabitEthernet0/0/0.71
encapsulation dot1Q 71
ip address 10.20.2.1 255.255.255.128
!
interface GigabitEthernet0/0/0.72
encapsulation dot1Q 72
ip address 10.20.2.129 255.255.255.128
!
interface GigabitEthernet0/0/1
ip address 192.168.2.2 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 0.0.0.0 0.0.0.0 192.168.2.1
ip route 10.10.0.0 255.255.0.0 10.255.0.1
ip route 10.10.5.0 255.255.255.0 10.255.0.1
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
```

!
!
end

BRANCH-RTR-01#

BRANCH-RTR-01#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 192.168.2.1 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 10 subnets, 6 masks
S 10.10.0.0/16 [1/0] via 10.255.0.1
S 10.10.5.0/24 [1/0] via 10.255.0.1
C 10.20.0.0/23 is directly connected, GigabitEthernet0/0/0.70
L 10.20.0.1/32 is directly connected, GigabitEthernet0/0/0.70
C 10.20.2.0/25 is directly connected, GigabitEthernet0/0/0.71
L 10.20.2.1/32 is directly connected, GigabitEthernet0/0/0.71
C 10.20.2.128/25 is directly connected, GigabitEthernet0/0/0.72
L 10.20.2.129/32 is directly connected, GigabitEthernet0/0/0.72
C 10.255.0.0/30 is directly connected, Tunnel0
L 10.255.0.2/32 is directly connected, Tunnel0
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.2.0/30 is directly connected, GigabitEthernet0/0/1
L 192.168.2.2/32 is directly connected, GigabitEthernet0/0/1
S* 0.0.0.0/0 [1/0] via 192.168.2.1

BRANCH-RTR-01#

BRANCH-RTR-01#show interfaces tunnel0

Tunnel0 is up, line protocol is up (connected)

Hardware is Tunnel

Internet address is 10.255.0.2/30

MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,

reliability 255/255, txload 1/255, rxload 1/255

Encapsulation TUNNEL, loopback not set

Keepalive not set

Tunnel source 192.168.2.2 (GigabitEthernet0/0/1), destination 10.0.0.1

Tunnel protocol/transport GRE/IP

Key disabled, sequencing disabled

Checksumming of packets disabled

Tunnel TTL 255

Fast tunneling enabled

Tunnel transport MTU 1476 bytes

Tunnel transmit bandwidth 8000 (kbps)

Tunnel receive bandwidth 8000 (kbps)

Last input never, output never, output hang never

Last clearing of "show interface" counters never

Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1

Queueing strategy: fifo

Output queue: 0/0 (size/max)

5 minute input rate 51 bits/sec, 0 packets/sec

5 minute output rate 0 bits/sec, 0 packets/sec

98 packets input, 6272 bytes, 0 no buffer

Received 0 broadcasts, 0 runts, 0 giants, 0 throttles

0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort

0 input packets with dribble condition detected

0 packets output, 0 bytes, 0 underruns

0 output errors, 0 collisions, 0 interface resets

0 unknown protocol drops

0 output buffer failures, 0 output buffers swapped out

BRANCH-RTR-01#

DHCP_SERVER

DHCP-SVR

Physical
 Config
 Services
Desktop
 Programming
 Attributes

SERVICES

HTTP
 DHCP
 DHCPv6
 TFTP
 DNS
 SYSLOG
 AAA
 NTP
 EMAIL
 FTP
 IoT
 VM Management
 Radius EAP
 PRP

DHCP

Interface
 FastEthernet0
 Service
 ☒ On
 ☐ Off

Pool Name
 serverPool

Default Gateway
 0.0.0.0

DNS Server
 0.0.0.0

Start IP Address :
 172
 16
 0
 0

Subnet Mask:
 255
 255
 255
 0

Maximum Number of Users :
 512

TFTP Server:
 0.0.0.0

WLC Address:
 0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
VLAN90	10.10.2.2	172.16.0...	10.10.2....	255.255....	100	0.0.0.0	0.0.0.0
VLAN40	10.10.0.2	172.16.0...	10.10.0....	255.255....	100	0.0.0.0	0.0.0.0
VLAN50-ENGINEERING	10.10.4.1	172.16.0...	10.10.4....	255.255....	80	0.0.0.0	0.0.0.0
VLAN60-EXEC	10.10.7.1	172.16.0...	10.10.7....	255.255....	100	0.0.0.0	0.0.0.0
VLAN30-HR	10.10.6.1	172.16.0...	10.10.6....	255.255....	100	0.0.0.0	0.0.0.0
VLAN20-FINANCE	10.10.5.1	172.16.0...	10.10.5....	255.255....	100	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	172.16.0.0	255.255....	512	0.0.0.0	0.0.0.0

Finance (VLAN 20) to HR (VLAN 30)

```
C:\> ping 10.10.6.100

Pinging 10.10.6.100 with 32 bytes of data:

Reply from 10.10.6.100: bytes=32 time=1ms TTL=127
Reply from 10.10.6.100: bytes=32 time=12ms TTL=127
Reply from 10.10.6.100: bytes=32 time<1ms TTL=127
Reply from 10.10.6.100: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.6.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>|
```

Sales (VLAN 40) to Engineering (VLAN 50)

```
C:\>ping 10.10.4.100

Pinging 10.10.4.100 with 32 bytes of data:

Reply from 10.10.4.100: bytes=32 time<1ms TTL=127
Reply from 10.10.4.100: bytes=32 time<1ms TTL=127
Reply from 10.10.4.100: bytes=32 time=10ms TTL=127
Reply from 10.10.4.100: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.4.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>|
```

HQ Host to ISP Loopback (Internet Access)

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Request timed out.
Request timed out.
Reply from 8.8.8.8: bytes=32 time=10ms TTL=253

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 5ms

C:\>
```

Branch VLAN 70 to HQ VLAN 20 (Across GRE Tunnel)

```
C:\>ping 10.10.5.101

Pinging 10.10.5.101 with 32 bytes of data:

Reply from 10.10.5.101: bytes=32 time=11ms TTL=125
Reply from 10.10.5.101: bytes=32 time=12ms TTL=125
Reply from 10.10.5.101: bytes=32 time=11ms TTL=125
Reply from 10.10.5.101: bytes=32 time=12ms TTL=125

Ping statistics for 10.10.5.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 12ms, Average = 11ms

C:\>
```

Finance CANNOT Ping HR (After ACL Application)

```
C:\>ping 10.10.6.10

Pinging 10.10.6.10 with 32 bytes of data:

Reply from 10.10.5.1: Destination host unreachable.
Reply from 10.10.5.1: Destination host unreachable.
Reply from 10.10.5.1: Destination host unreachable.
Reply from 10.10.5.1: Destination host unreachable.

Ping statistics for 10.10.6.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

DHCP Lease Confirmation

```
0.0.0.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::204:9AFF:FE15:30CD
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 10.10.6.100
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   10.10.6.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>
```